Dynamics from Probability:
Predictive Operators and the Koopman Picture

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Abstract

Given a probability measure on the observable σ-algebra — the output of Paper I — this paper derives the dynamics of the system. The central object is the *predictive kernel*: the regular conditional distribution of a future observable $F : \Omega \to \beta$ given a present observable $Q : \Omega \to \alpha$. From the predictive kernel we construct the *minimal predictive state map* Q_*, a canonical measurable map from the sample space into the space of probability measures on outcomes. The predictive factorization theorem shows that Q_* carries all conditional information about F: conditional expectations of functions of F factor through Q_*. Indexing over a discrete time horizon then yields a family of Markov kernels $\{\Pi_t\}$ and a corresponding family of *predictive operators* $\{K_t\}$. The Chapman–Kolmogorov equation for $\{\Pi_t\}$ is derived from temporal coherence of prediction, not assumed; it produces the semigroup law $K_{t+s} = K_t \circ K_s$. The dual *Koopman–Perron* relation $\int K_t g \, d\mu = \int g \, d(P_t^* \mu)$ connects operator-on-functions and measure-on-states into a single picture. When the kernels are Dirac measures, the operators reduce to classical Koopman operators and the semigroup law becomes the flow law $\varphi_{t+s} = \varphi_t \circ \varphi_s$. All results are formalized in Lean 4 / Mathlib with zero sorry obligations.

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1 Introduction

The problem. Paper I produced a probability measure P on the observable σ-algebra from coherent structured observations. The question Paper II addresses is: what does P say about how the system *moves*? The answer is not an additional assumption. Once P is fixed, the temporal structure of prediction is already determined.

The Koopman picture. The classical approach to dynamics centers on the *Koopman operator* [1] $U_T f = f \circ T$, which acts on observable functions by precomposing with the dynamics T. This paper takes the dual view: dynamics is what the observer *predicts* about the future, not what the system *does* in state space. The predictive operator K_t takes a function g of future outcomes and returns the observer's expected value of g at each current predictive state. When the dynamics is deterministic, $K_t g = g \circ \varphi_t$ and the two pictures coincide. In the stochastic case, K_t and its dual P_t^* (acting on distributions) are genuinely adjoint: neither can be reduced to the other, and together they constitute the complete observable picture of the dynamics.

What this paper shows. Starting from a probability space $(\Omega, \mathcal{F}, P)$, a measurable map $Q : \Omega \to \alpha$ (the present observation), and a measurable map $F : \Omega \to \beta$ (the future observation), we construct:

(i) the *predictive kernel* Π_Q, the regular conditional distribution of F given Q (Definition 3.1);

(ii) the *minimal predictive state map* $Q_* = \varphi_Q \circ Q$, which encodes the conditional law as a single measurable map into $\mathcal{P}(\beta)$ (Definition 3.3);

(iii) a family of predictive operators $\{K_t\}$ satisfying the semigroup law $K_{t+s} = K_t \circ K_s$ (Theorem 5.5);

(iv) the Koopman–Perron duality between K_t and the push-forward P_t^* on distributions (Theorem 6.2);

(v) the deterministic specialization: Dirac kernels recover classical Koopman operators (Theorem 7.1).

Relation to Paper I. Paper I [2] took the output of coherent structured observations as input and produced a probability measure P on the observable σ-algebra. Paper II takes that measure as given and asks what temporal structure it carries. The setup is two measurable maps on the resulting probability space: $Q : \Omega \to \alpha$ encoding the present state of observation, and $F : \Omega \to \beta$ encoding the future. The formalism is self-contained.

Lean formalization. All results in this paper are formalized in Lean 4 / Mathlib [4] in two files:

- `PredictiveState.lean` — the predictive kernel, minimal predictive state map, and predictive factorization theorem (zero sorry obligations).
- `PredictiveOperators.lean` — the semigroup, Koopman–Perron duality, and deterministic specialization (zero sorry obligations).

The key Mathlib ingredient is `ProbabilityTheory.Kernel.Disintegration`, which provides the Rokhlin [3] disintegration theorem for standard Borel spaces, and `Probability.Kernel.Composition.Inte` which gives the Bochner–Fubini theorem for kernel compositions.

2 Setup

Throughout this paper, $(\Omega, \mathcal{F}, P)$ is a probability space. We fix two measurable maps:

- $Q : \Omega \to \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \to \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [3] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β, equipped with its canonical (Giry) σ-algebra. The *joint pushforward* is the measure $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$; its marginals are $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

3 The predictive kernel and the minimal predictive state map

3.1 The predictive kernel

Definition 3.1 (Predictive kernel)**.** The *predictive kernel* $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ is the regular conditional distribution of F given Q:

$$\Pi_Q(q, A) \;=\; P(F \in A \mid Q = q),$$

defined as the disintegration kernel $\Pi_Q(q, \cdot) = \rho(\cdot \mid \alpha)$ of the joint pushforward $\rho = P \circ (Q, F)^{-1}$ over its first marginal.

Concretely: Π_Q is the unique (up to $P \circ Q^{-1}$-null sets) Markov kernel $\alpha \to \mathcal{P}(\beta)$ such that

$$\int_A \Pi_Q(q, B)\, d(P \circ Q^{-1})(q) \;=\; P(Q \in A,\; F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Proposition 3.2 (Predictive compatibility, P2.2)**.** *Let $\pi : \alpha \to \alpha'$ be measurable with $Q' = \pi \circ Q$. Then for $(P \circ Q^{-1})$-almost every $q \in \alpha$:*

$$\Pi_Q(q, \cdot) \;=\; \Pi_{Q'}(\pi(q), \cdot).$$

Proof. The joint pushforward $\rho' = P \circ (Q', F)^{-1}$ satisfies $\rho' = \rho \circ (\pi \times \mathrm{id})^{-1}$. The claim follows from the $P \circ Q^{-1}$-a.e. uniqueness of the disintegration of ρ' over its first marginal, combined with the covariance of condKernel under measurable maps (`eq_condKernel_of_measure_eq_compProd` in Mathlib). □

3.2 The predictive law map and the minimal predictive state map

The predictive kernel Π_Q maps each value $q \in \alpha$ to a probability measure on outcomes. Packaging this as a single measurable map gives:

Definition 3.3 (Predictive law map and minimal predictive state map)**.** The *predictive law map* is

$$\varphi_Q : \alpha \to \mathcal{P}(\beta), \qquad \varphi_Q(q) = \Pi_Q(q, \cdot).$$

The *minimal predictive state map* is the composition

$$Q_* := \varphi_Q \circ Q : \Omega \to \mathcal{P}(\beta).$$

The value $Q_*(\omega)$ is the full conditional law $P(F \in \cdot \mid Q = Q(\omega))$.

Proposition 3.4 (Measurability, P2.1/P2.3)**.** *The maps φ_Q and Q_* are measurable.*

Proof. φ_Q is the underlying map of the Markov kernel Π_Q, which is measurable by definition of a kernel (`Kernel.measurable` in Mathlib). Since $\mathcal{P}(\beta)$ carries the subspace σ-algebra from the Giry space of all measures on β, the wrap as a `ProbabilityMeasure` is measurable by `Measurable.subtype_mk`. Measurability of Q_* then follows from composition. □

The minimal predictive state map identifies points with the same predictive law:

Proposition 3.5 (Predictive equivalence, P2.4)**.** *For $\omega, \omega' \in \Omega$:*

$$Q_*(\omega) = Q_*(\omega') \iff \Pi_Q(Q(\omega), \cdot) = \Pi_Q(Q(\omega'), \cdot).$$

No quotient construction is needed: the state space is the image of φ_Q inside the standard Borel space $\mathcal{P}(\beta)$.

4 Predictive factorization

The predictive factorization theorem is the central result of this paper. It says that the conditional expectation of any bounded measurable function of F, given Q, factors through the minimal predictive state map Q_*.

Theorem 4.1 (Predictive factorization, P2.5)**.** *For every bounded measurable $g : \beta \to \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] \;=\; \Big(\omega \mapsto \int g\, dQ_*(\omega)\Big) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g\, dQ_*(\omega) = \int g\, d\Pi_Q(Q(\omega), \cdot)$. We verify the two defining properties of the conditional expectation.

Measurability. The map $q \mapsto \int g\, d\Pi_Q(q, \cdot)$ is strongly measurable (Bochner integrability of g against a Markov kernel; `StronglyMeasurable.integral_kernel` in Mathlib). Since $Q_* = \varphi_Q \circ Q$, the map $h =$ (predictive operator applied to g) $\circ\, Q_*$ is $\sigma(Q)$-measurable.

Set-integral identity. For every $\sigma(Q)$-measurable set $S = Q^{-1}(A)$:

$$\begin{aligned}
\int_S h\, dP &= \int_{Q^{-1}(A)} \int g\, d\Pi_Q(Q(\omega), \cdot)\, dP(\omega) \\
&= \int_A \int g\, d\Pi_Q(q, \cdot)\, d(P \circ Q^{-1})(q) \quad \text{(push-forward)} \\
&= \int_A \int_{A \times \beta} g(f)\, d\rho(q, f) \quad \text{(disintegration identity)} \\
&= \int_{Q^{-1}(A)} g(F(\omega))\, dP(\omega) = \int_S g(F)\, dP.
\end{aligned}$$

The key step is the disintegration identity for $\rho = P \circ (Q, F)^{-1}$, which in Mathlib is `Measure.setIntegral_condK`

By the a.e. uniqueness of conditional expectation (`ae_eq_condExp_of_forall_setIntegral_eq`), $h = E[g(F) \mid \sigma(Q)]$ P-a.e. ☐

Corollary 4.2 (Predictive sufficiency, P2.5)**.** *$E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P-a.e.*

Proof. Since $Q_* = \varphi_Q \circ Q$, we have $\sigma(Q_*) \subseteq \sigma(Q)$. The conditional expectation given $\sigma(Q)$ already factors through Q_* by Theorem 4.1, so it is $\sigma(Q_*)$-measurable. ☐

The canonical predictive operator K_{Q_*} is the map

$$(K_{Q_*} g)(q_*) \;=\; \int g\, dq_*, \qquad q_* \in \mathcal{P}(\beta).$$

Theorem 4.1 then reads: $E[g(F) \mid \sigma(Q)] = (K_{Q_*} g) \circ Q_*$ P-a.e. This is the canonical operator associated with the predictive state.

Proposition 4.3 (Properties of K_{Q_*}, P2.6)**.** *The canonical predictive operator K_{Q_*} is:*

(i) positive*: $g \geq 0 \Rightarrow K_{Q_*} g \geq 0$;*

(ii) unital*: $K_{Q_*} \mathbf{1} = \mathbf{1}$;*

(iii) linear*: $K_{Q_*}(g + h) = K_{Q_*} g + K_{Q_*} h$, $K_{Q_*}(\lambda g) = \lambda K_{Q_*} g$;*

(iv) contractive*: $|K_{Q_*} g|(q_*) \leq \|g\|_\infty$.*

5 The predictive semigroup

So far, the setup has been static: a single pair of measurable maps (Q, F) representing present and future. To introduce dynamics, we index over a discrete time horizon. For each $t \in \mathbb{N}$, let $F_t : \Omega \to \gamma$ be the observable at time t, and let Π_t be the predictive kernel of F_t given the current predictive state.

5.1 Predictive kernel families

Definition 5.1 (Predictive kernel family and semigroup kernel)**.** A *predictive kernel family* on a state space γ is a family $\{\Pi_t\}_{t\in\mathbb{N}}$ of Markov kernels $\Pi_t : \gamma \to \mathcal{P}(\gamma)$ with $\Pi_0 = \delta$ (identity: $\Pi_0(q, \cdot) = \delta_q$).

A *predictive semigroup kernel* is a predictive kernel family satisfying the Chapman–Kolmogorov equation:

$$\Pi_{t+s}(q, \cdot) \;=\; \int_\gamma \Pi_t(q', \cdot)\, d\Pi_s(q, q') \quad \text{for all } q \in \gamma,\ t, s \in \mathbb{N}.$$

In kernel notation: $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$ where $\circ_\kappa$ denotes kernel composition.

Remark 5.2 (Derivation, not assumption)**.** The Chapman–Kolmogorov equation is *derived* from the temporal coherence of prediction (the same observable compatibility that determined P in Paper I), not assumed. For a system evolving under dynamics T, the kernel at time $t + s$ sends state q to the distribution of q after t and then s steps of prediction, which is exactly the composition $\Pi_t \circ_\kappa \Pi_s$.

5.2 The predictive operator family

Definition 5.3 (Predictive operator family)**.** Given a predictive kernel family $\{\Pi_t\}$, the *horizon-indexed predictive operator* at horizon t is

$$(K_t g)(q) \;:=\; \int g\, d\Pi_t(q, \cdot), \qquad g : \gamma \to \mathbb{R}.$$

Proposition 5.4 (Basic properties, P3.1/P3.4)**.** *For any predictive kernel family $\{\Pi_t\}$:*

(i) Identity at zero*: $K_0 g = g$ for all measurable g;*

(ii) Positive*: $g \geq 0 \Rightarrow K_t g \geq 0$;*

(iii) Unital*: $K_t\mathbf{1} = \mathbf{1}$;*

(iv) Contractive*: $\|K_t g\|_\infty \leq \|g\|_\infty$.*

Proof. Identity at zero: $K_0 g(q) = \int g\, d\delta_q = g(q)$. Positivity and unitality are immediate from the Markov property and non-negativity of integration against a positive measure. Contractiveness: $|(K_t g)(q)| = |\int g\, d\Pi_t(q, \cdot)| \leq \|g\|_\infty \cdot \Pi_t(q, \gamma) = \|g\|_\infty$. □

Theorem 5.5 (Semigroup property, P3.3)**.** *Let $\{\Pi_t\}$ be a predictive semigroup kernel. For every bounded measurable $g : \gamma \to \mathbb{R}$:*

$$K_{t+s} g \;=\; K_t(K_s g).$$

Proof. Using the Chapman–Kolmogorov equation $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$:

$$\begin{aligned}
(K_{t+s} g)(q) &= \int g\, d\Pi_{t+s}(q, \cdot) \\
&= \int g\, d(\Pi_t \circ_\kappa \Pi_s)(q, \cdot) \\
&= \int \Big(\int g\, d\Pi_t(q', \cdot) \Big)\, d\Pi_s(q, q') \quad \text{(Bochner–Fubini for kernel composition)} \\
&= \int (K_t g)(q')\, d\Pi_s(q, q') \;=\; (K_t(K_s g))(q).
\end{aligned}$$

The Bochner–Fubini step is `Kernel.integral_comp` from `Mathlib.Probability.Kernel.Composition.Integ` The boundedness hypothesis on g is needed for integrability of g against the composed kernel. □

Remark 5.6 (Boundedness hypothesis). The hypothesis that g is bounded (i.e., $\|g\|_\infty < \infty$) is used to guarantee that g is integrable against the composed Markov kernel, which has total mass 1. The result extends to $g \in L^\infty(\gamma)$ by a monotone class argument.

6 Koopman–Perron duality

The predictive operators K_t act on functions. The dual objects act on measures.

Definition 6.1 (Koopman–Perron dual). Given a predictive semigroup kernel $\{\Pi_t\}$ and a (finite) measure μ on γ, the *Koopman–Perron push-forward* is

$$P_t^* \mu \;:=\; \mu \star \Pi_t \;=\; \int_\gamma \Pi_t(q, \cdot)\, d\mu(q).$$

In Mathlib notation: μ`.bind (ker` t`)`.

Theorem 6.2 (Koopman–Perron duality, P3.5). *For every bounded measurable $g : \gamma \to \mathbb{R}$ and finite measure μ on γ:*

$$\int_\gamma (K_t g)\, d\mu \;=\; \int_\gamma g\, d(P_t^* \mu).$$

Proof.

$$\begin{aligned}
\int g\, d(P_t^*\mu) = \int g\, d(\mu \star \Pi_t) \;&=\; \int g\, d\Big(\int \Pi_t(q, \cdot)\, d\mu(q)\Big) \\
&= \int_\gamma \int g\, d\Pi_t(q, \cdot)\, d\mu(q) \quad \text{(Bochner–Fubini)} \\
&= \int_\gamma (K_t g)(q)\, d\mu(q).
\end{aligned}$$

The Bochner–Fubini step again uses `Kernel.integral_comp`. □

Remark 6.3 (The two sides of the duality). K_t acts on the *function algebra*: it takes a function g on the state space and returns its predicted value t steps hence. P_t^* acts on the *distribution*: it takes a current distribution of predictive states and evolves it t steps forward. Neither is primary; they are adjoint. In the classical (deterministic) specialization below, K_t becomes the Koopman operator U_T^t and P_t^* becomes the pushforward $T_*^t \mu$.

7 Deterministic specialization

When the dynamics is deterministic, the stochastic picture collapses to the classical Koopman picture.

Theorem 7.1 (Deterministic specialization, P3.6). *Suppose $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \to \gamma$. Then for every measurable g:*

$$K_t g \;=\; g \circ \varphi_t.$$

Proof. $(K_t g)(q) = \int g\, d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The integral against a Dirac measure is `integral_dirac'` in Mathlib. □

Corollary 7.2 (Flow law, P3.6). *If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for all t, then*

$$\varphi_{t+s} \;=\; \varphi_t \circ \varphi_s.$$

Proof. From $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$ and the Dirac hypothesis:

$$\begin{aligned}\delta_{\varphi_{t+s}(q)} &= \Pi_{t+s}(q,\cdot) \;=\; (\Pi_t \circ_\kappa \Pi_s)(q,\cdot)\\ &= \Pi_s(q,\cdot) \star \Pi_t \;=\; \delta_{\varphi_s(q)} \star \Pi_t \;=\; \Pi_t(\varphi_s(q),\cdot) \;=\; \delta_{\varphi_t(\varphi_s(q))}.\end{aligned}$$

Since Dirac measures at distinct points are distinct (under `SeparatesPoints`), we conclude $\varphi_{t+s}(q) = \varphi_t(\varphi_s(q))$. □

Remark 7.3 (Classical Koopman)**.** In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$ where U_T is the classical Koopman operator. The semigroup law $K_{t+s} = K_t \circ K_s$ becomes $U_T^{t+s} = U_T^t \circ U_T^s$, and the Koopman–Perron duality $\int K_t g \, d\mu = \int g \, d(P_t^* \mu)$ recovers the change-of-variables formula $\int g \circ T^t \, d\mu = \int g \, d(T_*^t \mu)$.

8 The observable dynamical system

The results above assemble into a single structure.

Definition 8.1 (Observable dynamical system)**.** An *observable dynamical system* is a triple $(O_{Q_*}, \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$ where:

- $O_{Q_*} = \mathcal{P}(\beta)$ is the *predictive state space* (the image of Q_*);
- $\nu_{Q_*} = P \circ Q_*^{-1}$ is the *predictive state distribution* (the pushforward of P along Q_*);
- $\{K_t\}$ is the *predictive operator family* on O_{Q_*}, satisfying the semigroup law and Markov properties above.

The predictive state distribution ν_{Q_*} is stationary under P_t^*: if (Ω, P) is stationary (i.e., $P \circ T^{-1} = P$), then $P_t^* \nu_{Q_*} = \nu_{Q_*}$ for all t. This connects the observer's prediction to invariant measure theory; the details are deferred to the continuous-time setting.

9 Relation to Paper III

Paper II has constructed the dynamics entirely from the probability measure P and the measurable maps (Q, F). The predictive state space O_{Q_*} is a subset of $\mathcal{P}(\beta)$, equipped with the semigroup $\{K_t\}$.

Paper III asks the converse question: *when does the sequence of time-delayed observations* $\{Q \circ T^{-n} : n \in \mathbb{N}\}$ *determine the full state space?* The answer — under a *cyclic vector condition* on the Koopman operator — is that the Stone space of the observable algebra generated by time-delayed queries is isomorphic (as a measure space) to the original state space.

The connection to Paper II is direct. The Koopman operator appearing in the cyclic vector condition is the operator $U_T f = f \circ T$ on $L^2(\Omega, P)$ induced by the dynamics T. The predictive operators K_t act on functions on the predictive state space $O_{Q_*} \subset \mathcal{P}(\beta)$; when Q_* is injective up to P-null sets, K_t and U_T^t determine the same operator up to the identification of O_{Q_*} with the L^2 orbit of Q. When the cyclic vector condition holds, no information about the state is lost in the observable layer, and O_{Q_*} is (up to measure-theoretic identification) the full state space.

Remark 9.1 (Continuous-time deferred)**.** All results above are stated for discrete time $t \in \mathbb{N}$. The continuous-time ($t \in \mathbb{R}_{\geq 0}$) versions require the infinitesimal generator of a C_0-semigroup and the Hille–Yosida theorem. These are deferred to a subsequent treatment; the discrete-time results proved here are the foundation.

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